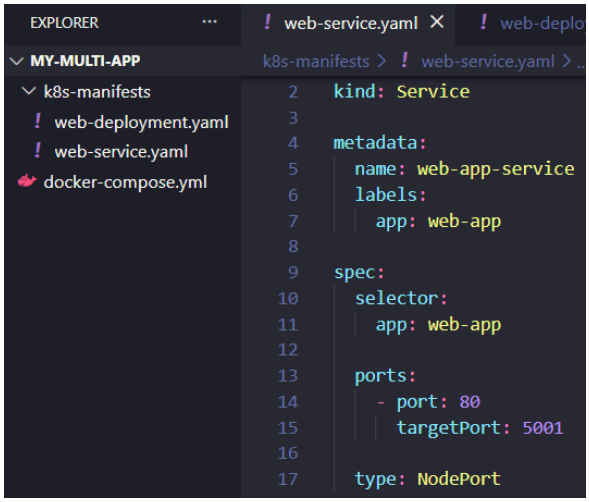
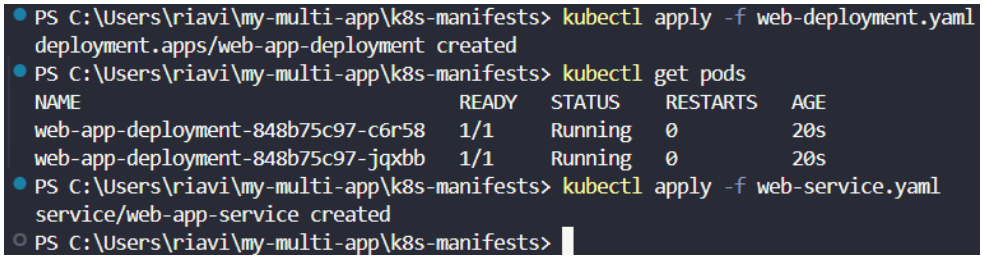
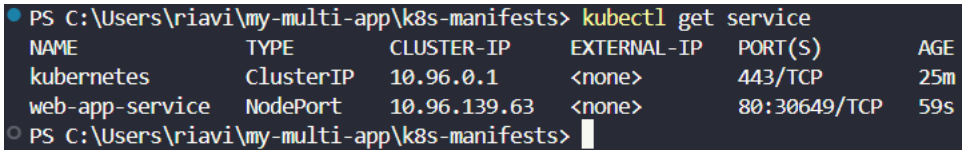
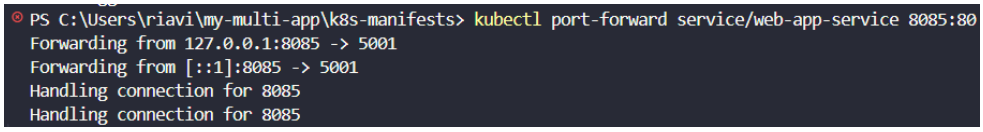
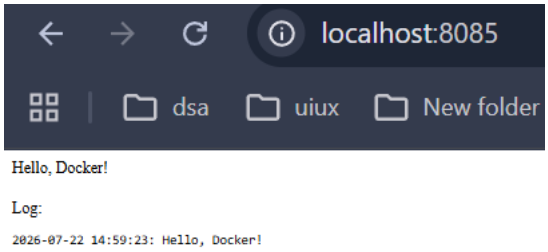
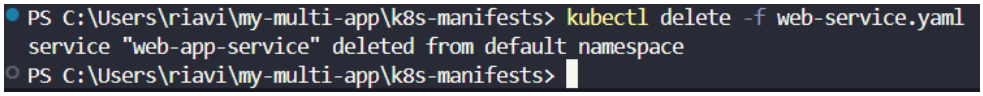
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Assignment No. 13

Subject:	DevOps Lab
TCODE:	TE7950
Name of Student	**Deepak Rathod**
PRN No.	**23070122175**
Branch	CSE, Batch (2023-27), L3
Academic Year & Semester	2026, Sem VII
Date of Performance	23rd July, 2026
Title of Assignment:	Kubernetes Service & Networking
Theory:	A Kubernetes Service is an abstraction that provides a stable network endpoint for accessing one or more Pods in a Kubernetes cluster. Since Pods are temporary and their IP addresses may change, a Service uses labels and selectors to route incoming traffic to the appropriate Pods. A NodePort Service exposes the application outside the cluster by allocating a port on each cluster node, allowing external access to the application. During local development, kubectl port-forward can also be used to securely forward traffic from a local machine to a Kubernetes Service for testing and verification. This approach simplifies communication between applications and ensures reliable access to containerized workloads. **Commands Used** • Created the Kubernetes Deployment: kubectl apply -f web-deployment.yaml • Verified that the Pods were running: kubectl get pods • Created the Kubernetes Service: kubectl apply -f web-service.yaml • Displayed the available Kubernetes Services: kubectl get service • Displayed detailed information about the Service: kubectl describe service web-app-service • Displayed the Service endpoints:

	kubectl get endpoints • Displayed the running Pods with labels: kubectl get pods --show-labels • Displayed the application logs: kubectl logs web-app-deployment-<pod-name> • Forwarded the Kubernetes Service to the local machine: kubectl port-forward service/web-app-service 8085:80 • Deleted the Kubernetes Service: kubectl delete -f web-service.yaml • Deleted the Kubernetes Deployment: kubectl delete -f web-deployment.yaml
Implementation:	**Step 1: Create the Kubernetes Service YAML**  Created Kubernetes Service Manifest — Created the web-service.yaml file defining a Kubernetes NodePort Service to expose the Flask web application running in the Kubernetes Pods. **Step 2: Apply the Service**  Applied Kubernetes Service — Applied the Kubernetes Service manifest using kubectl apply, creating a NodePort Service for the deployed Flask application. **Step 3: Find the Service Information** 

	Verified Kubernetes Service — Verified the Kubernetes NodePort Service and identified the assigned NodePort (30649) used to expose the Flask web application outside the Kubernetes cluster. **Step 4: Access the Application**   Accessed Application Through Kubernetes Service — Used kubectl port-forward to forward the Kubernetes Service to the local machine and successfully accessed the Flask web application at http://localhost:8085, verifying that it was running correctly. **Step 5: Clean Up**  Deleted Kubernetes Service — Removed the Kubernetes Service using the kubectl delete command, cleaning up the networking resources created for the application.

Subject Faculty Name: Prof. Arijit Dutta